

Joshua Aney

Rochester, MN | josh.aney@icloud.com | github.com/joshaney324 | joshaney324.github.io

EDUCATION

Montana State University

Bachelor of Science, Computer Science, GPA: 3.96

Bozeman, MT

May 2026

SKILLS & COURSEWORK

Languages: Python, MATLAB, Java, C, SQL, C#

Frameworks: PyTorch, TensorFlow, WebDataset, .NET Core, SLURM, NumPy, Matplotlib

Tools: Linux, JetBrains Products, GitHub, VSCode, Adobe Illustrator, Overleaf, Claude Code

EXPERIENCE

Research Technologist

June 2026 - Current, (Intern Summers 2024, 2025)

CaMP Lab (Cybernetics and Motor Physiology), Mayo Clinic

Rochester, MN

- **Tools:** MATLAB, Signal Processing, Adobe Illustrator, Linear Algebra, Technical Writing
- Performed computational motor and language mapping on epilepsy patients to localize task-active brain regions.
- Recorded sEEG experiments in the Epilepsy Monitoring Unit and preprocessed raw BCI2000 data for analysis.
- Developed a novel method for dynamic brain network identification from evoked responses.
- Modified Canonical Response Parameterization for better temporal feature extraction from auditory evoked responses.

Undergraduate Research Assistant

January 2026 - June 2026

HACR Lab (Harnessing Automation in Cybersecurity Reasoning), Montana State University

Bozeman, MT

- **Tools:** Pytorch, Ghidra, High Performance Computing, SLURM
- Built pipelines to produce control flow graphs from compiled binaries for code embedding generation.
- Designed PyTorch Graph Neural Network architectures to leverage the features of control flow graphs.
- Ran machine learning workloads on Tempest, MSU's high performance computing system.

Teaching and Curriculum Development

Fall 2025, Spring 2026

Montana State University

Bozeman, MT

- **Tools:** Java, HTMX, SQLite, HTML, CSS
- Co-designed a semester-long web development course project template with a professor and three students.
- Implemented core functionality (MVC controllers, auth, multi-step workflows) for a veterinary clinic system.
- Structured a progressive assignment sequence so students could add controllers from established codebase patterns.
- Assisted students in learning core data structures algorithms (trees, hashmaps, depth first search, etc.)
- Taught students how to debug in their IDE's.

TECHNICAL PROJECTS

Machine Learning and AI Models from Scratch | *Python, NumPy, Linear Algebra, Latex*

2024-2026

- Implemented from scratch: Supervised, unsupervised, and reinforcement learning algorithms (Naïve Bayes, KNN, K-Means, DBSCAN, Multi-layer Perceptron, Q-Learning, SARSA, Value Iteration) and optimizers (backpropagation, genetic algorithm, differential evolution, particle swarm optimization, simulated annealing).
- Implemented a First-Order-Logic engine to perform inference on knowledge bases using resolution.

Bozeman Bike Trails Application | *C#, Razor Pages, SQL Server, Mapbox*

2024

- Developed a web application to track user-ridden bike trail GPS data

Hand Tracking / ASL Alphabet Detection | *Python, TensorFlow, MediaPipe*

2024

- Created a pipeline to collect data, train a dense neural network, and provide real time predictions for video inputs.

EXTRACURRICULAR ACTIVITIES & LEADERSHIP

President

Montana State University Tennis Club

August 2023 – Present

Bozeman, MT

Section Leader

Montana State University Drumline

April 2025 – January 2026

Bozeman, MT